

1. Introduction

2. Related Work

3. Observation Model

Let $\mathcal{X} \subset \mathbb{R}^{n_w}$ denote the space of fixed-length observation windows, where n_w is the window length. In the experiments of Sections 8–9 these observations are vibration windows, but the formulation does not depend on this particular sensing modality.

Let $\theta \in \Theta$ denote the latent physical state of the system. In a rolling-element bearing application, examples of θ include nominal operation, outer-race degradation, inner-race degradation, ball damage, or transitional states not yet assigned a semantic label.

Let $\eta \in \mathcal{E}$ denote nuisance variables that modify the observed signal without changing the underlying physical state. Typical nuisance variables include:

- amplitude scaling due to gain or load variation;
- phase shift due to sampling alignment;
- speed perturbation;
- DC offset or baseline drift;
- sensor placement and mounting effects;
- additive measurement noise.

The observation process is written as

$$x = F(\theta, \eta), \quad x \in \mathcal{X}. \quad (1)$$

Physically, nuisance variables are constrained to bounded operating ranges: amplitude scaling, phase and speed perturbation, and offset all vary within finite intervals dictated by the installation and the operating envelope. We therefore record the following standing assumption, which is used in Section 4 to establish boundedness of the nominal functional structure.

Assumption 1 (Bounded nuisance range). *The nuisance parameter space $\mathcal{E}$ is a compact subset of a finite-dimensional parameter space, and for the nominal state θ_N the map $\eta \mapsto F(\theta_N, \eta)$ is uniformly bounded and equicontinuous on $\mathcal{E}$.*

The diagnostic objective is not to identify η , nor is it necessarily to reconstruct the full forward model F , which may be unknown, nonlinear, high-dimensional, or impractical to identify. The diagnostic objective is to recover information about θ from x while remaining invariant to nuisance degrees of freedom. This produces the inverse problem

$$x \mapsto \theta, \quad (2)$$

or, more precisely, recovery of the physical state modulo nuisance transformations of the observation process. The inverse problem (2) is ill-posed in the classical sense [5, 12]: distinct nuisance configurations produce distinct observations corresponding to the same latent state, so the forward map is not injective in the relevant sense and no stable pointwise inverse exists without regularization.

4. Quotient-Space Formulation

The key mathematical object is the equivalence relation induced by nuisance transformations. Let $\mathcal{T}$ denote the physically motivated family of nuisance transformations introduced in Section 3: amplitude scaling, phase shift, speed perturbation, offset, and additive noise.

For the formal quotient argument, we let G_{nuis} denote the transformation group generated by $\mathcal{T}$, together with the identity transformation and the inverses of its elements. This is a closure construction: even if the physically motivated transformations in $\mathcal{T}$ are realized only over bounded operational ranges, the group generated by transformations of the same type is mathematically well defined.

The formal quotient-space construction is therefore stated in terms of the group action of G_{nuis} on $\mathcal{X}$. The compactness argument below does *not* require the full group G_{nuis} to be compact; it uses only the compact nuisance parameter set $\mathcal{E}$ of Assumption 1. Thus the mathematical quotient is defined using a genuine group action, while any empirical instantiation samples a compact, physically admissible subset of nuisance configurations. The consequences of this gap are revisited in Section 11.

Definition 1 (Nuisance equivalence). *Two observations $x_1, x_2 \in \mathcal{X}$ are nuisance-equivalent, written $x_1 \sim x_2$, if there exists $g \in G_{\text{nuis}}$ such that*

$$x_2 = g \cdot x_1, \quad (3)$$

and both observations correspond to the same latent physical state θ .

The orbit of an observation x under nuisance transformations is

$$[x] = \{g \cdot x : g \in G_{\text{nuis}}\}, \quad (4)$$

and the quotient space is the set of such equivalence classes:

$$\mathcal{X}/G_{\text{nuis}} = \{[x] : x \in \mathcal{X}\}. \quad (5)$$

We write $\pi : \mathcal{X} \rightarrow \mathcal{X}/G_{\text{nuis}}$ for the canonical projection $\pi(x) = [x]$, and equip $\mathcal{X}/G_{\text{nuis}}$ with the quotient topology, so that π is continuous by construction.

Definition 2 (Nominal quotient manifold). *The nominal quotient manifold $\mathcal{M}_{\text{N}}$ is the subset of $\mathcal{X}/G_{\text{nuis}}$ generated by observations sharing the nominal latent physical state:*

$$\mathcal{M}_{\text{N}} = \{[x] \in \mathcal{X}/G_{\text{nuis}} : x = F(\theta_{\text{N}}, \eta), \eta \in \mathcal{E}\}. \quad (6)$$

Proposition 1 (Compactness of the nominal nuisance orbit). *Under Assumption 1, the nominal nuisance orbit*

$$\mathcal{O}_{\text{N}} = \{F(\theta_{\text{N}}, \eta) : \eta \in \mathcal{E}\} \subset \mathcal{X}$$

is relatively compact: its closure $\overline{\mathcal{O}_{\text{N}}}$ is a compact subset of $\mathcal{X}$, where $\mathcal{X}$ is regarded as a space of fixed-length observation windows equipped with the uniform (sup-norm) topology.

Proof. By Assumption 1, the family $\{F(\theta_{\text{N}}, \eta)\}_{\eta \in \mathcal{E}}$ is uniformly bounded and equicontinuous in the sup-norm topology. By the Arzelà–Ascoli theorem, such a family is relatively compact in that topology. Therefore $\overline{\mathcal{O}_{\text{N}}}$ is compact. $\square$

Corollary 1 (Boundedness and connectedness of the nominal quotient structure). *Assume the hypotheses of Proposition 1. Since the quotient projection π is continuous, the image $\pi(\overline{\mathcal{O}_{\text{N}}})$ is compact in $\mathcal{X}/G_{\text{nuis}}$. If, in addition, $\mathcal{E}$ is connected and $\eta \mapsto F(\theta_{\text{N}}, \eta)$ is continuous, then $\mathcal{O}_{\text{N}}$ is connected and hence so is $\pi(\mathcal{O}_{\text{N}}) = \mathcal{M}_{\text{N}}$. Consequently $\mathcal{M}_{\text{N}}$ is contained in a compact, connected subset of the quotient space.*

Proof. Continuous images of compact sets are compact and continuous images of connected sets are connected; $\mathcal{E}$ is a product of bounded intervals in the instantiation of Section 8, hence connected, and $\mathcal{M}_{\text{N}} = \pi(\mathcal{O}_{\text{N}})$ by Definition 2. $\square$

Remark 1 (On the use of the term “manifold”). *Proposition 1 and Corollary 1 establish compactness and connectedness properties of the nominal structure under explicit assumptions, but they do not establish a smooth or topological manifold structure in the differential-geometric sense, such as local Euclidean charts, a well-defined dimension, or curvature. The term functional manifold is used here, consistently with the companion report on manifold saturation [4] and with common usage in the representation-learning literature under the manifold hypothesis [1], to denote a bounded, connected region of the quotient space generated by a fixed latent regime under nuisance variability. A fully differential-geometric treatment is left to future work.*

Proposition 2 (Quotient factorization of diagnostic maps). *Assume that G_{nuis} acts on $\mathcal{X}$, that the action preserves the latent physical state, and that a diagnostic map $h : \mathcal{X} \rightarrow \mathcal{Y}$ depends only on that latent physical state. Then h is constant on G_{nuis} -orbits and therefore factors through the quotient projection π ; that is, there exists $\bar{h} : \mathcal{X}/G_{\text{nuis}} \rightarrow \mathcal{Y}$ such that*

$$h = \bar{h} \circ \pi. \quad (7)$$

Proof. Let $x_1, x_2 \in \mathcal{X}$ lie in the same G_{nuis} -orbit, so that $x_2 = g \cdot x_1$ for some $g \in G_{\text{nuis}}$. By assumption the action of g preserves the latent physical state, and h depends only on that state; hence $h(x_1) = h(x_2)$. Therefore h is constant on equivalence classes, and by the universal property of quotient maps it factors through π , yielding $\bar{h}$ with $h = \bar{h} \circ \pi$. $\square$

Corollary 2 (Equivalence-class diagnosis). *If two observations lie in the same nuisance orbit, no physically meaningful diagnostic decision distinguishes them on the basis of nuisance variation alone. Consequently, nominal observations should be identified as elements of a nominal equivalence class, or of a connected region of such classes, in the quotient space.*

Under this formulation, anomaly detection becomes a membership problem:

$$[x] \in \mathcal{M}_N \quad \text{or} \quad [x] \notin \mathcal{M}_N. \quad (8)$$

An anomaly is therefore not primarily a label. It is a failure of membership in the nominal quotient manifold. Classification, when required, is a subsequent operation that assigns non-nominal observations to other quotient-space regions or fault manifolds; this is developed in Section 6.

Remark 2 (Relation to the functional-topological foundation). *The compact, connected structure $\mathcal{M}_N$ obtained in Corollary 1 plays the role of the compact perceptual manifold of finite Hausdorff radius postulated in the functional-topological foundation of Di Santi [3], and the present section can be read as supplying one mechanism by which such a structure arises for a specific physical process. The two formal settings are, however, not identical and should not be conflated. In Di Santi [3], compactness is asserted directly for the set of valid realizations of a physical process, viewed as a subset of a Banach space of signals. Here, compactness is established for the nominal orbit in observation space (Proposition 1) and then transported to the quotient $\mathcal{X}/G_{\text{nuis}}$ by continuity of π (Corollary 1). The quotient construction is what makes the resulting structure low-complexity: nuisance directions are collapsed rather than merely bounded. Accordingly, the present work does not re-derive the general functional-topological claim; it specializes it to a nuisance group acting on a rotating-machinery observation process and supplies the additional geometric object, the quotient, that the diagnostic decision actually requires.*

5. Functional Manifolds and Blueprint Saturation

Section 4 established that the nominal structure is compact and connected. This section explains why that structure is the object a representation learner can plausibly recover, and makes the saturation criterion of Di Santi [4] precise enough to be certified rather than merely observed.

5.1. Functional manifolds as quotient geometry

The quotient-space formulation supplies a mechanism through which functional manifolds arise. A functional manifold is not postulated as an arbitrary structure that an algorithm happens to discover. It arises because nuisance transformations generate orbits in observation space, and the quotient projection collapses each orbit to a single equivalence class.

The observed structure therefore has two distinct sources:

- (i) physical continuity of the underlying system, which makes the orbit of a fixed latent regime a connected set; and
- (ii) identification of observations that differ only by nuisance transformations, which removes the directions along which that orbit is extended.

In raw signal space, nominal operation may appear as a broad and noisy cloud, because the nuisance directions inflate it in every coordinate that the sensor happens to be sensitive to. In quotient space those directions are collapsed, and what remains is closer to the structure of the latent physical state. This is the reason a representation that is insensitive to nuisance variability can expose compact functional regions where raw observations show none. The projection does not create physical structure; it removes variability that is irrelevant to the latent state.

This also clarifies the relation between anomaly detection and classification. A supervised classifier learns a partition of labelled observations. A quotient-space diagnostic operator learns a representation in which physically equivalent observations are identified. Operational labels may subsequently be attached to the recovered regions, but the regions precede the labels; that is the subject of Section 6.

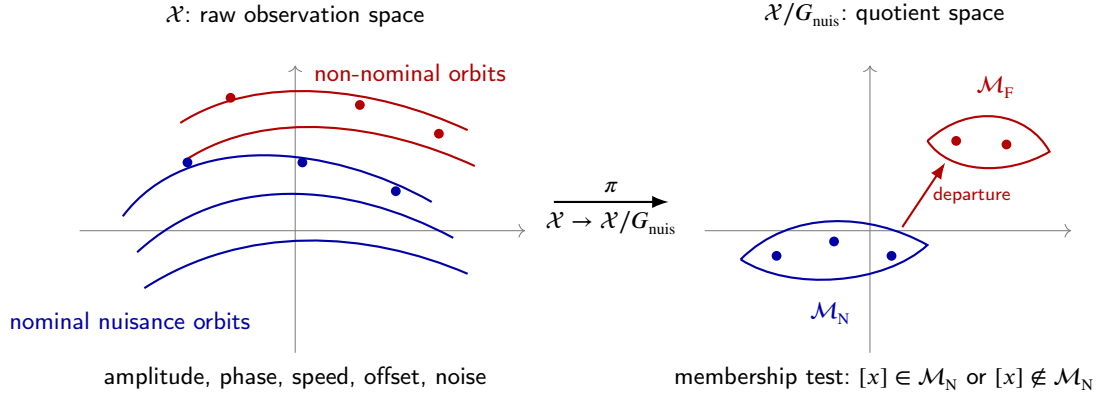


Figure 1: Functional manifolds as quotient geometry. In raw observation space, nuisance transformations generate extended orbits of observations that share the same latent physical state. The quotient projection π collapses those orbits into equivalence classes, exposing compact nominal and non-nominal functional regions. Diagnosis is then a membership test in $\mathcal{M}_N$ rather than a partition of labelled observations.

5.2. The saturation radius

The companion report [4] proposes that a regime becomes learnable when its geometric descriptors stabilize under additional observations, written informally as

$$H(n) \rightarrow H^*, \quad r_{\max}(n) \rightarrow r^*, \quad V(n) \rightarrow V^*, \quad (9)$$

for an entropy, a radius and a volume descriptor. As stated, (9) is a description of observed behaviour rather than a claim with a definite content: neither the estimators nor the mode of convergence is specified. We therefore fix one descriptor precisely, the radius, and show that doing so brings saturation within reach of the same machinery used in Section 7.

Let $\phi : \mathcal{X} \rightarrow \mathcal{Z}$ be a representation map and let ν denote the law induced on $\mathcal{Z}$ by drawing η from the nominal nuisance law and applying $\phi \circ F(\theta_N, \cdot)$. Fix a reference point $c \in \mathcal{Z}$, for instance the mean of ν , and define the scalar

$$D = d(\phi(x), c), \quad x = F(\theta_N, \eta), \quad (10)$$

with distribution function F_D .

Definition 3 (Population and empirical saturation radius). *For $p \in (0, 1)$ the population p -radius of the nominal regime is the quantile*

$$R_p = F_D^{-1}(p) = \inf \{r \geq 0 : F_D(r) \geq p\}, \quad (11)$$

and given n nominal observations with radial values $D_1, \dots, D_n$ the empirical p -radius $\hat{R}_p(n)$ is the corresponding empirical quantile. The descriptor $r_{\max}(n)$ of (9) is the case $p \rightarrow 1$, and the p_{95} and p_{98} radii used in the companion work are the cases $p = 0.95$ and $p = 0.98$.

The first thing this definition buys is that the object being estimated exists and is finite, which does not follow from (9) on its own but does follow from Section 4.

Corollary 3 (Finiteness of the saturation radius). *Under Assumption 1, if ϕ is continuous then D is a bounded random variable and $R_p < \infty$ for every $p \in (0, 1)$; moreover $\sup_{\eta \in \mathcal{E}} D < \infty$, so the regime has a finite radius in the Hausdorff sense.*

Proof. By Proposition 1 the closure $\overline{\mathcal{O}_N}$ is compact, and ϕ is continuous, so $\phi(\overline{\mathcal{O}_N})$ is compact in $\mathcal{Z}$. The map $z \mapsto d(z, c)$ is continuous, and a continuous real function on a compact set is bounded and attains its supremum. Hence D is bounded and every quantile of F_D is finite. $\square$

Corollary 3 is the precise sense in which the compactness result of Section 4 underwrites the finite-Hausdorff-radius premise of the functional-topological foundation discussed in Remark 2. Saturation can then be stated as a convergence claim about an estimator of a finite quantity.

Proposition 3 (Consistency of the empirical saturation radius). *Let $D_1, D_2, \dots$ be independent draws of (10) and suppose F_D is strictly increasing in a neighbourhood of R_p , so that the quantile is unique. Then*

$$\hat{R}_p(n) \longrightarrow R_p \quad \text{almost surely as } n \rightarrow \infty. \quad (12)$$

Proof. By the Glivenko–Cantelli theorem the empirical distribution function converges to F_D uniformly and almost surely. Uniqueness of the quantile at level p transfers uniform convergence of the distribution function to pointwise convergence of the quantile function at p ; see, for example, Serfling [10, Sec. 2.3] or van der Vaart [13, Lemma 21.2]. $\square$

Proposition 3 makes (9) a theorem rather than an observation, and it identifies the limit as a specific population quantile. It is, however, purely asymptotic, and therefore silent on the question the operational criterion actually poses, namely how many observations suffice.

5.3. Finite-sample certification of saturation

That question is answered by the argument of Section 7.4. Definition 3 makes the saturation radius a quantile of a nominal distribution estimated from finitely many samples, which is exactly what the local operational radius is; only the space differs, distances in $\mathcal{Z}$ rather than residuals in $\mathbb{R}$. Proposition 4 is stated for an arbitrary exchangeable scalar sequence and therefore applies unchanged.

We note the forward dependency explicitly. Proposition 4 is proved in Section 7.4 rather than here because its operational role is the calibration of the decision boundary, which is where a reader will expect to find it; but it is a statement about exchangeable real-valued sequences and is logically independent of everything specific to residuals, as recorded in Remark 14. The corollary below therefore borrows a later result without circularity.

Corollary 4 (Distribution-free certification of the saturation radius). *Let $D_1, \dots, D_n, D_{n+1}$ be exchangeable with an atomless marginal, let $\alpha = 1 - p$, and set*

$$\hat{R}_p^\star = D_{(k_\alpha)}, \quad k_\alpha = \lceil p(n+1) \rceil. \quad (13)$$

Then, provided $n \geq 1/\alpha - 1$,

$$p \leq \mathbb{P}(D_{n+1} \leq \hat{R}_p^\star) = \frac{k_\alpha}{n+1} \leq p + \frac{1}{n+1}, \quad (14)$$

and $\hat{R}_p^\star$ equals the largest observed radial value $D_{(n)}$ precisely when $1/\alpha - 1 \leq n < 2/\alpha - 1$.

Proof. Apply Proposition 4, Corollary 6 and Corollary 7 with r_i replaced by D_i . Neither the coverage argument nor the two corollaries uses any property of the scalar sequence beyond exchangeability and atomlessness. $\square$

Corollary 4 unifies the two halves of the framework, and its practical content is the same in both. Certifying that a regime’s empirical extent covers a fraction p of its nominal orbit requires at least $1/\alpha - 1$ observations, so 49 at $p = 0.98$; and because the certified radius is the observed maximum throughout $1/\alpha - 1 \leq n < 2/\alpha - 1$, a saturation curve that appears to have levelled off anywhere in that band has levelled off at the most variable statistic available, which is a property of the estimator rather than evidence about the regime. For $p = 0.98$ a non-degenerate saturation radius requires $n \geq 99$.

Remark 3 (Plug-in reference points break exchangeability). *The radial variable (10) is defined against a fixed reference c . In practice c is estimated from data, and the companion work additionally estimates the projection in which distances are measured from the same sample. Both substitutions make $D_1, \dots, D_n$ depend on one another and destroy the exchangeability of the augmented sequence, so Corollary 4 then holds only approximately, with an error that vanishes as the reference is estimated more precisely. The remedy is the same split-sample construction available in Remark 12: estimate the reference point, and the projection if one is used, on a block of nominal observations disjoint from the block used to compute the radius. This restores exchangeability by construction and costs only observations, not assumptions.*

Remark 4 (What a saturation curve does and does not establish). *Observed flattening of $\hat{R}_p(n)$ is a stopping heuristic, not a proof that R_p has been recovered, exactly as convergence of the commissioning criterion is not a proof that the population residual quantile has been recovered (Remark 15). An upper quantile estimated from few upper-tail observations can remain flat over an interval and then move again when a further extreme nominal observation arrives, and Corollary 4 says nothing to the contrary: it bounds coverage at a given n , not the increments of the curve. Claims of the form “the geometry has stabilized by $n \approx 500$ ” should therefore be read as descriptions of the observed trajectory, with the coverage statement supplying the part that is guaranteed.*

Remark 5 (The saturation curve and the guarantee pull in opposite directions). *The measurements reported in Section 9 sharpen Remark 4 into a concrete warning about the estimator, and the effect is not small. Inside the degenerate range the certified radius is the sample maximum, so it drifts systematically upward with n : on simulated nominal data at $p = 0.98$ its mean rises from about 2% above R_p at $n = 49$ to about 6% above at $n = 98$, and then falls discontinuously at $n = 99$, where an interior order statistic first becomes available. A saturation curve computed with the certified estimator therefore cannot flatten inside the band; apparent settling there is the maximum of a growing sample.*

The interpolated estimator behaves in exactly the complementary way. It produces the familiar smooth curve approaching R_p from below, and it undercovers at every sample size examined. The reassuring visual and the coverage guarantee are thus in tension: the estimator that makes a saturation plot look convergent is precisely the one that fails to certify, and the estimator that certifies does not produce a flat curve until the degenerate range has been left behind. Saturation should accordingly be reported with the attained coverage alongside the curve, and with $n \geq 2/\alpha - 1$.

Remark 6 (The saturation radius is a diagnostic, not a detector). *One further empirical observation limits what a saturation analysis can be asked to show. Measuring the certified radius of each simulated regime under a shared two-dimensional nominal projection separates them barely at all: the ratios of fault to nominal radius are between 0.98 and 1.04, that is, no separation. The reason is mechanical. A low-dimensional principal projection of a raw vibration window is dominated by the shaft harmonics, whereas the fault signature is a low-amplitude impulse train contributing almost nothing to the leading variance directions. The same simulator, the same nominal-only fitting and the same membership rule yield 99–100% detection when membership is tested through a reconstruction residual instead (Section 9).*

Two consequences follow. First, a saturation plot is evidence about the sampling of the nominal geometry, not about regime separability, and should not be cited for the latter. Second, this is the empirical reason the framework poses diagnosis through a residual, as in Section 7, rather than through a radius in a projection: Assumption 2 asks the score to track distance to the nominal manifold, and a leading-variance radius demonstrably does not.

6. Emergent Classification via Expert Feedback

Sections 4–5 concern a single regime: the nominal one. This section states what the same construction gives when several regimes have been recovered, which is the point at which the framework makes contact with classification. The answer is deliberately asymmetric: the geometry supplies a great deal about when a class should *not* be rejected, and very little about when a competing class should be, and that asymmetry is the substance of the section.

6.1. Two levels of classification

Classification in this framework occurs at two levels, and conflating them is the source of most of the confusion the inverse-problem view is meant to dispel.

The first is *structural* classification. A new observation is assigned to a recovered functional region, to an emerging region that has not yet saturated, to a transition between regions, or to none of them. This assignment requires no semantic labels at all: it is induced by the geometry of the quotient space and by the membership test (8).

The second is *operational* classification. Once a region has stabilized in the sense of Section 5, an expert or a downstream policy may attach meaning to it: normal operation, harmless variation, friction, heat, degradation, or a known fault regime. Classification is therefore not absent before labels exist; it first appears as geometric membership and only later becomes semantic.

6.2. Manifold-level labelling

Let $\{\mathcal{M}_j\}_{j=1}^k$ denote a family of recovered compact regions in $\mathcal{X}/G_{\text{nuis}}$, of which $\mathcal{M}_N$ is one. Labels are attached at the level of regions rather than of samples, through a map

$$L : \{\mathcal{M}_j\}_{j=1}^k \longrightarrow \mathcal{Y}, \quad L(\mathcal{M}_j) = y_j. \quad (15)$$

The contrast with supervised practice is the object being labelled. A supervised classifier consumes $x_i \mapsto y_i$ over many individual samples; here the expert supplies $\mathcal{M}_j \mapsto y_j$, once per recovered region. The labelling effort is proportional to the number of regimes, not to the number of observations.

Given a residual r_j for each region, in the sense of Assumption 2, and a radius τ_j calibrated for that region by Proposition 4, the admissible set for an observation is

$$\Gamma(x) = \{j \in \{1, \dots, k\} : r_j(x) \leq \tau_j\}, \quad (16)$$

and the reported diagnosis is

$$\hat{y}(x) = \begin{cases} L(\mathcal{M}_j), & \Gamma(x) = \{j\}, \\ \{L(\mathcal{M}_j)\}_{j \in \Gamma(x)}, & |\Gamma(x)| > 1 \text{ (ambiguous)}, \\ \text{unknown}, & \Gamma(x) = \emptyset. \end{cases} \quad (17)$$

Writing the rule as a set rather than an arg min is not a stylistic choice. It is what the calibration of Section 7 actually licenses, and it makes the two failure modes explicit rather than hiding them behind a nearest-region assignment: an observation may satisfy no region, which is the abstention (8) already anticipates, or it may satisfy several. The construction is the class-conditional, or Mondrian, form of a conformal predictor [11, 14], arrived at here from the quotient geometry rather than imported.

Corollary 5 (Per-class non-rejection). *Fix a region $\mathcal{M}_j$ and suppose the residuals r_j evaluated on observations from regime j satisfy Assumption 3, with τ_j calibrated from $n_j \geq 1/\alpha_j - 1$ such observations by (28). Then for a further observation x drawn from regime j ,*

$$\mathbb{P}(j \in \Gamma(x)) \geq 1 - \alpha_j. \quad (18)$$

Consequently the true regime is retained in the admissible set with probability at least $1 - \alpha_j$, simultaneously for every regime, without any assumption on the residual distributions and without any calibration data from competing regimes.

Proof. Apply Proposition 4 to the exchangeable sequence generated by r_j on regime j ; the event $\{j \in \Gamma(x)\}$ is precisely $\{r_j(x) \leq \tau_j\}$. The statement is per-regime and the regimes are calibrated independently, so no joint assumption is required. $\square$

Corollary 5 formalizes what the framework buys. Each region is calibrated only against its own nominal behaviour, so a new regime can be added after deployment without recalibrating the others, and a regime for which no data yet exists simply contributes nothing to $\Gamma(x)$. This is the sense in which the taxonomy may grow: adding $\mathcal{M}_{k+1}$ changes (16) by one term and leaves (18) untouched for $j \leq k$.

Several structural consequences follow, and they are the ones the companion report anticipates informally [4]: two regions may carry the same label when they are operational variants of one regime; a single nominal label may resolve into several regions when the underlying behaviour is heterogeneous; an unfamiliar observation need not be forced into the nearest known class; and expert effort attaches to regions rather than to samples.

6.3. What coverage does not give: separation

Corollary 5 bounds the probability of wrongly *excluding* the true regime. It says nothing whatever about wrongly *including* a competing one, and the distinction is not a technicality.

Remark 7 (Coverage is not discrimination). *Nothing in (18) prevents $|\Gamma(x)| > 1$. The acceptance regions $\{r_j \leq \tau_j\}$ are calibrated independently and may overlap arbitrarily; in the extreme, if two regions are geometrically indistinguishable under the available residuals, every observation from either is admitted by both, and (18) still holds exactly. A guarantee about non-rejection is therefore compatible with a diagnosis that is always ambiguous. Reducing $|\Gamma(x)|$ requires separation between regions, which is a property of the representation and must be established separately; it does not follow from the quotient construction of Section 4, from compactness, or from any calibration argument.*

Remark 8 (Two obstructions visible in the present construction). *The computational realization of Section 7 does not by itself supply that separation, for two distinct reasons, and the experiments of Section 9 exhibit both.*

First, a scalar residual is a many-to-one map. The rule (24) tests departure from the nominal region and reports a single number, from which the direction of departure cannot be recovered. Detection of a fault and identification of its type are therefore not the same problem in this construction, and the second requires k residuals, one per region, each with its own representation and its own calibration, not one residual with k thresholds.

Second, the low-dimensional radial diagnostic of Section 5 does not separate the regimes it describes. Measured on simulated data under a shared two-dimensional nominal projection, the ratio of fault radius to nominal radius lies between 0.98 and 1.04 (Section 9), that is, no separation at all, even though the same simulator and the same nominal-only fitting give 99–100% detection through the reconstruction residual. Saturation of a region and separability of that region from its neighbours are independent properties, and evidence for the former is not evidence for the latter.

This is the precise sense in which fault classification remains open in the present work, and it locates the gap left by the companion technical note [2] rather than closing it. What has been established is that diagnosis factors through the quotient (Proposition 2), so a correct classifier exists as a function of $[x]$ alone; that regions, once recovered, can be labelled and calibrated individually with distribution-free per-class validity (Corollary 5); and that abstention and ambiguity are reported rather than concealed (17). What has not been established is that any particular representation renders the non-nominal regions separable. Section 12 records the experiment that would test it.

7. Computational Realization and Local Calibration

Sections 3–6 characterize the diagnostic object: a compact, connected nominal region $\mathcal{M}_N$ in a quotient space, with anomaly defined as failure of membership in that region. That characterization is not yet an algorithm. Two quantities must be supplied before (8) can be evaluated on data: a computable surrogate for distance to $\mathcal{M}_N$, and a decision boundary on that surrogate.

This section supplies both. Section 7.1 introduces the representation operator and the residual used as a distance surrogate. Section 7.2 explains, as a consequence of Proposition 2, why the representation is transferable across assets while the numerical scale of the residual is not, which is the structural reason a single global decision boundary cannot be used. Sections 7.3 and 7.4 then give the calibration procedure and its guarantee: the central new result, Proposition 4, shows that a radius chosen as an appropriate order statistic of nominal commissioning residuals attains a prescribed coverage level without any assumption on the residual distribution, and Corollary 6 converts this into a minimum commissioning length.

Throughout this section we index physical assets, or more generally independently commissioned monitoring environments, by k .

7.1. Representation operator and diagnostic residual

Let $E : \mathcal{X} \rightarrow \mathcal{Z}$ be an encoder and $D : \mathcal{Z} \rightarrow \mathcal{X}$ a decoder, with reconstruction

$$\hat{x} = D(E(x)), \quad (19)$$

and define the *diagnostic residual*

$$r(x) = \frac{1}{n_w} \|x - \hat{x}\|_2^2. \quad (20)$$

In the present realization E and D are the encoder and decoder of a one-dimensional convolutional autoencoder trained exclusively on nominal windows generated by Monte Carlo sampling of $\mathcal{E}$; autoencoders have long been used both as nonlinear representation learners [6] and as reconstruction-based anomaly detectors [9]. No non-nominal observation is used at any stage of representation learning.

Assumption 2 (Residual–distance proxy). *The diagnostic residual $r(x)$ is treated as a computable proxy that is monotonically related to distance to the nominal quotient manifold:*

$$r(x) \approx d([x], \mathcal{M}_N). \quad (21)$$

Assumption 2 is an operational hypothesis, not a consequence of the training objective. Minimizing mean-squared reconstruction error on nominal data does not by itself guarantee that r is isometric, or even monotonic, with respect to any particular metric on $\mathcal{X}/G_{\text{nuis}}$. The assumption is therefore evaluated empirically in Section 9, where the separation between nominal and non-nominal residual distributions is the relevant evidence, and it is listed among the limitations in Section 11.

The substantive claim is not that mean-squared error is the correct residual. It is that a residual can serve as an operational test for membership in the nominal quotient manifold. Any score with the qualitative property in (21) may be substituted: a latent likelihood, a Mahalanobis distance in $\mathcal{Z}$, an energy score, a diffusion or denoising score, a predictive error, or a physics-informed residual. The calibration theory of Section 7.4 is stated for an arbitrary residual and is therefore unaffected by that choice.

7.2. Why the representation transfers but its scale does not

Proposition 2 states that the ideal diagnostic map factors through the quotient. Write that ideal map as the membership predicate

$$h(x) = \mathbf{1}[\bar{h}(x) \notin \mathcal{M}_N], \quad (22)$$

so that $h = \bar{h} \circ \pi$ by Proposition 2, and h is constant on nuisance orbits. The computable surrogate actually evaluated on data is

$$h_\tau(x) = \mathbf{1}[r(x) > \tau]. \quad (23)$$

Remark 9 (Invariance is inherited by the predicate, not by the score). *The surrogate (23) does not inherit the orbit-invariance of (22). The residual r is a function on $\mathcal{X}$, not on $\mathcal{X}/G_{\text{nuis}}$: it is not G_{nuis} -invariant, because the reconstruction error attained by a fixed pair (E, D) varies over an orbit even when the latent state does not. Consequently, for a fixed τ the level set $\{r = \tau\}$ is not the π -preimage of a subset of the quotient, and h_τ is not constant on orbits. What the quotient structure does guarantee, via Proposition 2, is that a correct diagnostic decision exists as a function of $[x]$ alone. Recovering it from r requires choosing τ so that the level set approximates $\partial\mathcal{M}_N$ under the nuisance law actually realized by the asset being monitored.*

Remark 9 has a direct architectural consequence, and it is the reason the calibration problem is posed per-asset rather than globally. The representation (E, D) encodes which directions of variation are nuisance and which are diagnostic; that information is a property of the asset family and is transferable. The numerical value of r additionally reflects the sensor coupling, mounting, operating point, preprocessing, and the particular optimization trajectory that produced (E, D) ; none of that is transferable. We therefore adopt the following separation:

- one representation operator, learned once from simulated nominal observations and thereafter held fixed;
- one scalar *local operational radius* τ_k^* per monitored asset, estimated from that asset's own nominal observations.

The decision rule for asset k at time t is then

$$\hat{y}(x_{k,t}) = \begin{cases} \text{anomalous,} & r(x_{k,t}) > \tau_k^*, \\ \text{nominal,} & r(x_{k,t}) \leq \tau_k^*. \end{cases} \quad (24)$$

This prediction is falsifiable and is tested directly in Section 9: transferring the residual *scale* across the simulation-to-field boundary should fail, while transferring the representation and re-estimating only τ_k^* should not.

7.3. Monte Carlo reference calibration

A reference boundary can be computed entirely in simulation. Let μ_{MC} and σ_{MC} denote the mean and standard deviation of r over held-out nominal Monte Carlo windows, and let $q \in (0, 1)$ denote a target nominal coverage level. A parametric reference is

$$\tau_{\text{MC}}(q) = \mu_{\text{MC}} + z_q \sigma_{\text{MC}}, \quad (25)$$

with z_q the corresponding standard-normal quantile; alternatively $\tau_{\text{MC}}(q)$ may be taken as the empirical q -quantile of the synthetic nominal residual distribution.

The level q is not part of the mathematical formulation. It is an operational risk parameter: lower values increase sensitivity at the cost of false alarms, higher values reduce false alarms at the cost of delayed detection of weak or incipient deviations.

Remark 10 (τ_{MC} is a reference, not a deployable radius). *By Remark 9, $\tau_{\text{MC}}(q)$ calibrates the residual scale of the simulator, not of any physical asset. It is reported here because it is the natural baseline and because its failure is informative, not because it is proposed as an operating point. The corresponding experiment is the direct-transfer condition of Section 9.*

7.4. Distribution-free calibration of the local operational radius

We now give the calibration procedure a guarantee. The construction is a distribution-free one-sided tolerance limit, in the sense introduced by Wilks [15] and developed in the modern conformal-prediction literature [7, 8, 14]. Its relevance here is that the local operational radius $\tau_k^\star$ is *exactly* such a tolerance limit on the nominal residual distribution of asset k , a correspondence that supplies both a coverage statement and a minimum commissioning length.

During commissioning, asset k acquires a finite sequence of observations believed to be nominal, and the shared representation converts them into residuals

$$r_1, r_2, \dots, r_n, \quad r_i = r(x_{k,i}). \quad (26)$$

Let $r_{(1)} \leq r_{(2)} \leq \dots \leq r_{(n)}$ denote the corresponding order statistics, and let r_{n+1} denote the residual of the next nominal observation from the same asset, that is, the first observation to be evaluated under the frozen radius.

Assumption 3 (Exchangeability of nominal commissioning residuals). *The residuals $r_1, \dots, r_n, r_{n+1}$ arising from nominal operation of asset k are exchangeable: their joint distribution is invariant under permutation of the indices. In addition, their common marginal distribution is atomless, so that ties occur with probability zero.*

Exchangeability is strictly weaker than independence and does not require the nominal residual distribution to be known, parametric, or even identified. It does require that commissioning and the immediately subsequent monitoring period be drawn from the same nominal regime; it is therefore violated by drift, by a non-stationary run-in transient, or by an asset that is already degraded at installation. These are precisely the failure modes recorded in Section 11, and Remark 13 below identifies the experiment that probes the assumption empirically.

Proposition 4 (Distribution-free coverage of the local operational radius). *Let Assumption 3 hold, let $\alpha \in (0, 1)$ be a target miscoverage level, and suppose the commissioning length satisfies*

$$n \geq \frac{1}{\alpha} - 1. \quad (27)$$

Define the local operational radius as the order statistic

$$\tau_k^\star = r_{(k_\alpha)}, \quad k_\alpha = \lceil (1 - \alpha)(n + 1) \rceil. \quad (28)$$

Then $k_\alpha \leq n$, so (28) is well defined, and

$$1 - \alpha \leq \mathbb{P}(r_{n+1} \leq \tau_k^\star) = \frac{k_\alpha}{n + 1} \leq 1 - \alpha + \frac{1}{n + 1}. \quad (29)$$

In particular the false-alarm probability on the next nominal observation satisfies $\mathbb{P}(r_{n+1} > \tau_k^\star) \leq \alpha$.

Proof. We first verify feasibility. Since n is an integer, $\lceil (1 - \alpha)(n + 1) \rceil \leq n$ holds if and only if $(1 - \alpha)(n + 1) \leq n$, which rearranges to $1 \leq \alpha(n + 1)$, that is, to (27). Hence $k_\alpha \in \{1, \dots, n\}$ and $r_{(k_\alpha)}$ is a well-defined order statistic of the commissioning sample.

Define the rank of the future residual within the augmented sample,

$$R = 1 + \#\{i \in \{1, \dots, n\} : r_i < r_{n+1}\}. \quad (30)$$

By Assumption 3 the marginal is atomless, so almost surely the $n + 1$ residuals are distinct and R is the position of r_{n+1} in the increasing ordering of $r_1, \dots, r_{n+1}$.

We claim that, almost surely,

$$\{r_{n+1} \leq r_{(k_\alpha)}\} = \{R \leq k_\alpha\}. \quad (31)$$

Indeed, $r_{(k_\alpha)} < r_{n+1}$ holds if and only if at least k_α of the first n residuals are strictly smaller than r_{n+1} , since $r_{(k_\alpha)}$ is the k_α -th smallest of those n values. By (30) this is equivalent to $R \geq k_\alpha + 1$. Taking complements gives (31).

It remains to compute the law of R . Exchangeability of $(r_1, \dots, r_{n+1})$ implies that the joint distribution is invariant under any permutation of the indices; combined with almost-sure distinctness, this implies that r_{n+1} is equally likely to occupy any of the $n + 1$ positions in the increasing ordering. Hence R is uniformly distributed on $\{1, \dots, n + 1\}$ and

$$\mathbb{P}(r_{n+1} \leq \tau_k^\star) = \mathbb{P}(R \leq k_\alpha) = \frac{k_\alpha}{n + 1}. \quad (32)$$

Finally, the definition of the ceiling gives $(1 - \alpha)(n + 1) \leq k_\alpha < (1 - \alpha)(n + 1) + 1$, and dividing by $n + 1$ yields the two outer inequalities in (29). $\square$

Three features of Proposition 4 matter for the application. First, no assumption is placed on the shape of the nominal residual distribution, which is essential here because that distribution is the output of a learned nonlinear operator and is neither Gaussian nor otherwise tractable. Second, the guarantee is per-asset and requires only that asset's own nominal data, which is what makes the shared-representation architecture of Section 7.2 admissible. Third, the coverage is two-sided: (29) bounds the attained level from above as well as below, so the procedure is not needlessly conservative, with the excess vanishing at rate $1/(n+1)$.

Corollary 6 (Minimum commissioning length). *Under Assumption 3, attaining distribution-free coverage $1 - \alpha$ by (28) requires*

$$n \geq \left\lceil \frac{1}{\alpha} \right\rceil - 1. \quad (33)$$

For the target coverage $1 - \alpha = 0.98$ used throughout this work, the minimum commissioning length is $n \geq 49$ observations, and (28) selects $k_\alpha = 49$ at $n = 49$, that is, the sample maximum.

Proof. Immediate from (27): if $n < 1/\alpha - 1$ then $k_\alpha > n$ and no order statistic of the commissioning sample attains the required level. Since n is an integer the condition is $n \geq \lceil 1/\alpha \rceil - 1$. Substituting $\alpha = 0.02$ gives $n \geq 49$, and $k_\alpha = \lceil 0.98 \cdot 50 \rceil = 49$. $\square$

Corollary 6 is worth emphasizing because it is a hard information-theoretic floor rather than a tuning choice: below 49 nominal observations, a 98% nominal-coverage radius cannot be certified from that asset's data alone by any distribution-free procedure, irrespective of the estimator used.

The floor also has a consequence that is easy to miss and that turns out to matter more in practice than the floor itself.

Corollary 7 (Degenerate range: the certified radius is the sample maximum). *Under the hypotheses of Proposition 4, $k_\alpha = n$, and hence $\tau_k^* = r_{(n)} = \max_i r_i$, if and only if*

$$\frac{1}{\alpha} - 1 \leq n < \frac{2}{\alpha} - 1. \quad (34)$$

At $\alpha = 0.02$ this is $49 \leq n \leq 98$; an interior order statistic is obtained only from $n \geq 99$.

Proof. $k_\alpha = \lceil (1 - \alpha)(n + 1) \rceil = n$ holds if and only if $n - 1 < (1 - \alpha)(n + 1) \leq n$. The right inequality rearranges to $1 \leq \alpha(n + 1)$, that is $n \geq 1/\alpha - 1$, which is (27). The left inequality rearranges to $\alpha(n + 1) < 2$, that is $n < 2/\alpha - 1$. $\square$

Corollary 7 says that for every commissioning length in a band roughly twice the width of the floor, the certified radius is the sample maximum, which is the order statistic of largest variance. The practical implication is stronger than the floor: satisfying (33) is necessary but not sufficient for a stable radius, and a commissioning length chosen just above the floor purchases coverage at the cost of maximal estimator variance. Monte Carlo verification across six residual distributions (Section 9) gives a standard deviation of τ_k^* that is essentially flat over $n \in [49, 98]$, falls by a factor of about 1.6 at the first interior order statistic $n = 99$, and by a factor of about 2.5 by $n = 200$. For a target coverage of 98% the recommendation is therefore $n \geq 99$, not $n \geq 49$.

Remark 11 (Interpolated quantiles are anti-conservative). *Standard empirical-quantile routines return a value obtained by linear interpolation between adjacent order statistics, which for an upper quantile lies below the order statistic $r_{(k_\alpha)}$ prescribed by (28). Such an estimate therefore does not satisfy (29), and the resulting false-alarm rate is systematically above the nominal level α . This is a concrete, testable prediction: any implementation that calibrates with an interpolated p_{98} rather than with the order statistic $r_{(k_\alpha)}$ should exhibit a positive bias in false-alarm rate across all assets, in the same direction in every case. Monte Carlo evaluation over six residual distributions (Section 9) confirms this: the interpolated estimator loses 1.9 percentage points of coverage relative to $r_{(k_\alpha)}$, consistently in sign and magnitude and irrespective of the distribution, whereas the certified estimator attains the coverage $50/51 = 0.9804$ predicted at $n = 50$ to within Monte Carlo error. Of the two bias sources considered here, interpolation is by an order of magnitude the dominant one (cf. Remark 12). The correction implied by Proposition 4 is to replace the interpolated quantile by $r_{(k_\alpha)}$ and to enforce (33).*

Remark 12 (Data-dependent stopping breaks exactness, but is conservative here). *Proposition 4 treats n as fixed in advance. The stopping rule of Section 7.5 makes n a function of the observed residual sequence, and an adaptively chosen stopping time destroys the exchangeability of the augmented sample $(r_1, \dots, r_n, r_{n+1})$, so the coverage statement holds approximately rather than exactly. Two distinct effects must be separated, and they act in opposite directions.*

The first is a genuine selection effect: a rule that terminates when the upper-tail estimate has been locally stable tends to stop at instants of quiet tail behaviour, which biases $r_{(k_a)}$ downward and is therefore anti-conservative. The second is a sample-size effect: the rule does not stop at the floor but typically well above it, and by Corollary 7 the certified radius is the sample maximum throughout the relevant range, so a larger n yields a larger radius and higher coverage.

Monte Carlo evaluation (Section 9) shows the second effect dominates. Comparing the adaptive rule against a fixed- n calibration at the same realized n , which isolates selection from sample size, gives a coverage difference of at most 0.2 percentage points with inconsistent sign across residual distributions; the selection bias is therefore negligible at this configuration. Comparing the adaptive rule against a fixed $n = 50$ instead gives a coverage gain of about 0.6 percentage points, attributable to the larger realized n . Under the configuration used here, adaptive stopping is thus mildly conservative rather than anti-conservative. This should not be read as a general guarantee, since it depends on the stopping parameters and on operating inside the degenerate range of Corollary 7. Two remedies restore exactness if required: fix n in advance subject to (33), or use the stopping rule only to declare readiness and compute τ_k^ from a disjoint block of nominal observations.*

Remark 13 (The robustness ablation is an exchangeability probe). *Assumption 3 is testable in a specific sense. If the nominal commissioning residuals are exchangeable, then permuting their arrival order is distributionally inconsequential, and the calibrated radius, the convergence point, and the post-calibration false-alarm rate should vary only by finite-sample noise. The repeated-commissioning experiment of Section 9, in which the shared representation and all calibration parameters are held fixed while the commissioning observations and their order are permuted, is therefore not merely a robustness check: it is the natural empirical probe of the one assumption on which Proposition 4 rests. Systematic dependence on arrival order would falsify exchangeability and indicate drift within the commissioning window.*

Remark 14 (Scope: the proposition is not about residuals). *Nothing in Proposition 4 or in Corollaries 6 and 7 uses any property of r beyond exchangeability and atomlessness of the scalar sequence it generates. The results are therefore statements about exchangeable real-valued sequences, and they apply equally to any other nominal quantity estimated by an upper quantile. This is what licenses Corollary 4 of Section 5.3, where the same argument is applied to radial distances in representation space rather than to residuals, yielding the same floor and the same degenerate range. The saturation radius of a functional regime and the local operational radius of an asset are one estimation problem posed in two spaces.*

7.5. Convergence-based stopping and freezing

Corollary 6 gives a lower bound on the commissioning length but not a stopping time. In deployment the relevant question is when the estimated radius has stabilized enough to be frozen. Let

$$\tau_k(t) = \hat{Q}_{k,t}(1 - \alpha) \quad (35)$$

denote the estimate after t commissioning residuals, where $\hat{Q}_{k,t}$ is the empirical quantile function of $r_1, \dots, r_t$; by Proposition 4 the certified choice is $\hat{Q}_{k,t}(1 - \alpha) = r_{(\lceil(1-\alpha)(t+1)\rceil)}$. Define the relative variation over a look-back interval L ,

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t - L)|}{\max\{|\tau_k(t - L)|, \epsilon\}}, \quad (36)$$

with $\epsilon > 0$ a numerical stabiliser. Commissioning terminates at the first $t \geq N_{\min}$ for which $C_k(t) < \epsilon_C$ holds for m consecutive updates, at which point $\tau_k^* = \tau_k(t)$ is frozen and monitoring under (24) begins. If no such t is reached before a maximum budget $N_{\max}$, the asset is assigned a calibration-failure status and is not released into active monitoring.

Two constraints link this rule to the theory of Section 7.4. First, $N_{\min}$ must satisfy (33), since otherwise a converged radius can be declared at a sample size at which the target coverage is not certifiable; by Corollary 6 this requires

$N_{\min} \geq 49$ at $\alpha = 0.02$, and strictly greater than 49 if the degenerate sample-maximum case is to be avoided. Second, by Remark 12 the rule introduces a downward bias in τ_k^* , so the parameters $(N_{\min}, N_{\max}, L, \varepsilon_C, m)$ define a trade-off between commissioning duration and the size of that bias rather than a consistency property.

Remark 15 (What the stopping rule does and does not claim). *Convergence of (36) is not evidence that the population $(1 - \alpha)$ -quantile has been recovered. It is an operational statement that the estimate has been stable over the configured look-back under the configured tolerance. Non-convergence within $N_{\max}$ is correspondingly diagnostic rather than merely inconvenient: it indicates insufficient nominal data, non-stationary nominal operation, or startup behaviour inconsistent with the assumed nominal state, that is, a failure of Assumption 3. Distinguishing among these causes lies outside the present evaluation.*

7.6. Summary of the procedure

The realization consists of an offline stage performed once per asset family and an online stage performed once per asset.

Offline (once per asset family).

1. Sample nuisance configurations $\eta_i \in \mathcal{E}$ and generate nominal observations $x_i = F(\theta_N, \eta_i)$, thereby approximating the nominal orbit $\mathcal{O}_N$ of Proposition 1.
2. Train (E, D) on these nominal observations only, and freeze it.
3. Optionally record the synthetic reference $\tau_{MC}(q)$ of (25) as a baseline, subject to Remark 10.

Online (once per asset, at commissioning).

1. Acquire nominal residuals $r_1, r_2, \dots$ using the frozen (E, D) .
2. Update $\tau_k(t)$ by (35) and monitor $C_k(t)$ of (36), with $N_{\min}$ satisfying (33).
3. On convergence, freeze τ_k^* and begin monitoring under (24); on exhaustion of $N_{\max}$, withhold the asset with calibration-failure status.

No fault labels, no field representation learning, and no manual threshold selection occur at any point. The only asset-specific quantity is the scalar τ_k^* , and by Proposition 4 it carries a distribution-free coverage guarantee.

8. Experimental Design

The experiments are organized by the kind of evidence they supply rather than by increasing realism, because the three kinds are not interchangeable and a reader is entitled to know which claim rests on which.

The first kind is *verification*. Propositions 4 and 3 and their corollaries are mathematical statements; the appropriate check is whether the stated identities hold, not whether a dataset behaves well. These experiments use no data at all, and one of them uses no sampling either.

The second kind is *controlled instantiation*. A physics-motivated simulator supplies a setting in which the latent regime, the nuisance family and the departures from nominal are all known by construction, so the end-to-end chain of Section 7 can be exercised against ground truth.

The third kind is *field transfer*. Two public bearing benchmarks test whether a representation learned entirely in simulation retains diagnostic meaning on real measurements: the Case Western Reserve University (CWRU) dataset under controlled operating-condition variability, and the NASA Intelligent Maintenance Systems (IMS) run-to-failure dataset under naturally evolving degradation.

8.1. Design invariants

Four constraints hold throughout and are stated once here rather than repeated per experiment.

1. **No fault observation is used for fitting or calibration.** Fault data enters only at evaluation. This applies to the representation operator, to the reference threshold τ_{MC} , and to every operational radius.

2. **No field observation is used for representation learning.** The shared representation is trained on simulated nominal windows and transferred unchanged; CWRU and NASA IMS observations are used only to estimate a scalar radius and to evaluate.
3. **Calibration observations are disjoint from evaluation observations.** Reported false-alarm rates are computed on nominal observations not used to compute the radius, so coverage is an out-of-sample quantity.
4. **Every reported quantity is seeded.** Sampling, permutation and model initialization are all deterministic given the seed recorded with the results.

8.2. Verification of the calibration theory

Four checks, in decreasing order of strength.

Exhaustive enumeration. For small n the coverage identity of Proposition 4 is evaluated by enumerating all $(n + 1)!$ orderings of $n + 1$ distinct exchangeable values and counting the fraction in which the last falls at or below $r_{(k_\alpha)}$. This is a check of the proposition itself rather than of a Monte Carlo approximation to it, and it is run over a grid of α .

Heterogeneous residual families. Proposition 4 is distribution-free, so the appropriate empirical test is not the residual distribution of one trained model but a family chosen to be as dissimilar as possible: lognormal, gamma, exponential, Pareto, half-normal and a bimodal mixture, spanning light and heavy tails, bounded and unbounded support, and unimodal and bimodal shapes. Any systematic dependence of attained coverage on the member of the family would falsify the proposition.

The estimator–stopping design. The two conventions of Section 7.4 are crossed with two stopping rules, giving four arms:

	adaptive stopping ($N_{\min} = 40$)	fixed $n = 50$
interpolated p_{98}	A1	A3
certified $r_{(k_\alpha)}$	A2	A4

A1 reproduces the published pipeline exactly and A4 is the configuration to which Proposition 4 applies without qualification. Comparing the four cells separates the contribution of the estimator from that of the stopping rule, which the published configuration confounds. Because the A2–A4 contrast still mixes the stopping *selection* effect with the larger sample size the adaptive rule collects, a further control compares the adaptive rule against a fixed- n calibration on an independent block of the *same* realized size.

The exchangeability test. Assumption 3 is probed by the permutation test described in Remark 13, whose statistic is the absolute Spearman correlation between residual value and arrival index. Its size is measured under exchangeable streams and its power under an injected monotone drift of increasing magnitude, before it is applied to any real commissioning stream: a test with inflated size would make the robustness ablation look like evidence of a violated assumption, and a test without power would make it uninformative.

8.3. Controlled instantiation: the bearing simulator

The simulator generates windows of $n_w = 1200$ samples at $f_s = 12$ kHz, corresponding to 100 ms, with a nominal shaft frequency of 30 Hz. The nominal signal is a five-term harmonic expansion with geometric decay 0.55, plus a damped structural resonance at 3 kHz and broadband noise. It is not offered as a high-fidelity twin of any particular assembly; its purpose is to provide a regime with known latent structure, controlled nuisance variability and controlled departures.

Nuisance variables are drawn independently per window: amplitude $a \sim U(0.85, 1.15)$, phase $\phi \sim U(-0.3, 0.3)$, speed $s \sim U(0.95, 1.05)$, offset $\sim U(-0.05, 0.05)$, additional noise $\sim U(0, 0.03)$ and a short local dropout. These instantiate the compact nuisance set $\mathcal{E}$ of Assumption 1. Non-nominal regimes add impulse trains at 108s, 162s and 72s Hz for outer-race, inner-race and ball faults respectively, the last with a low-frequency amplitude modulation.

Three measurements are taken in this setting. The end-to-end chain of Section 7 is exercised with a representation operator fitted on nominal windows only, verifying that nominal residuals concentrate, that fault residuals do not, that

a single membership test detects every fault family, and that assets differing only by an installation signature identify materially different radii. The saturation radius of Section 5 is measured against a large-sample reference R_p , with its dispersion, its relative bias and its attained coverage reported as functions of n across and beyond the degenerate range. Finally the same radius is measured per regime under a shared nominal projection, to test whether saturation of a region implies separability from its neighbours.

8.4. Field transfer: CWRU

The shared representation is transferred unchanged to the CWRU Drive-End dataset, sampled at 12 kHz, so that the 1200-sample window again corresponds to 100 ms. Four operating conditions are treated as independent deployment environments: 1730, 1750, 1772 and 1797 RPM, comprising 1,414 healthy, 1,621 ball-fault, 1,621 inner-race-fault and 2,846 outer-race-fault windows.

Three calibration strategies are compared under the identical representation, so that only the information available for estimating the radius changes:

1. **Direct threshold transfer**, applying the simulator-derived τ_{MC} unchanged. This is the empirical test of Remark 9: the representation is predicted to transfer and its numerical scale is not.
2. **Field oracle**, computing the radius retrospectively from the complete healthy record of each condition. Operationally unavailable; included as an upper reference.
3. **Online calibration**, estimating the radius sequentially under the stopping rule of Section 7.5.

A separate reference is provided by independently trained asset-specific detectors, one representation and one preprocessing transformation fitted per condition on its complete healthy record. This reference is deliberately favourable and operationally unrealistic; its role is to bound what is lost by sharing one representation.

The initialization-robustness ablation permutes the commissioning observations and their arrival order while holding the representation and all calibration parameters fixed. Following Remark 13, this is reported not merely as a robustness check but as the empirical probe of Assumption 3.

8.5. Field transfer: NASA IMS

A second representation is constructed for the IMS bearing family from simulated nominal observations matched to that dataset's sampling and operating characteristics, and transferred unchanged to each of four independently monitored bearings from Experiment 2, each comprising 984 chronologically ordered recordings at a 10-minute interval.

Each bearing estimates its radius from an early-life segment and freezes it. Departure is declared by a persistence rule over a fixed window of recordings, and the reported quantities are the number of observations to convergence, the spread of the converged radii across bearings, the healthy-period false-alarm rate, the time of first persistent departure, and the lead time to the documented end of the run.

Unlike CWRU, this setting has no discrete fault classes and no balanced evaluation set; what it tests is whether an independently commissioned radius remains diagnostically meaningful across a complete degradation trajectory without further adaptation.

8.6. Estimator conventions, and a caveat on the field results

One point must be declared here rather than allowed to surface in Section 9. The field experiments as conducted use the *interpolated* empirical p_{98} , with $N_{\min} = 40$, $N_{\max} = 200$, look-back $L = 10$, tolerance $\varepsilon_C = 0.01$ and $m = 10$ consecutive stable updates. In the terminology of Section 8.2 they are the A1 arm.

Two consequences follow, and both are consequences of results proved above rather than of any defect discovered in the data. First, by Remark 11 an interpolated upper quantile lies at or below $r_{(k_\alpha)}$ and therefore does not satisfy the coverage bound, so the field radii do not carry the guarantee of Proposition 4; the systematic direction of the resulting false-alarm excess is predicted, and is examined in Section 9. Second, $N_{\min} = 40$ lies below the commissioning floor $1/\alpha - 1 = 49$ of Corollary 6, and the commissioning lengths the rule actually returns lie in $[49, 98]$, which by Corollary 7 is precisely the range in which the certified radius degenerates to the sample maximum.

Claiming the guarantee in the field therefore requires re-running calibration with $r_{(k_\alpha)}$ and $N_{\min} \geq 2/\alpha - 1 = 99$, reporting both configurations. That comparison is inexpensive, since it operates on the residual streams alone and requires no retraining, and it is the natural extension of the present work; it is recorded in Section 12.

Remark 16 (A numerical coincidence that must not be read as evidence). *The commissioning lengths returned in the field cluster at 49, and the coverage floor of Corollary 6 at $\alpha = 0.02$ is also 49. These arise from unrelated mechanisms. The stopping rule cannot emit a calibration size below $\max\{N_{\min} - 1, \text{warm-up} - 1 + L\} + m$, which equals 49 at the configuration used and is independent of α ; the coverage floor is $1/\alpha - 1$ and does not involve the stopping parameters. The observed clustering is therefore uninformative about Corollary 6. Disentangling the two requires moving one floor without the other, for instance by setting $m = 5$, which moves the stopping floor to 44 while leaving α untouched.*

8.7. Declared uncertainties and reproducibility

Two details of the NASA IMS protocol are recorded as unresolved rather than silently fixed, because both affect reported lead times. The persistence rule appears in the source code with a threshold of three exceedances in five recordings, while the accompanying text describes eight in ten; and the recording-level score is computed as the mean of the window residuals in the code, while a 95th-percentile aggregation is also defined. The results in Section 9 state which convention was used.

The experiments divide cleanly by what a reader must obtain in order to repeat them. The verification and simulator experiments generate their own data from the simulator described in Section 8.3, which is distributed with the accompanying package; they therefore require no download, no dataset licence and no deep-learning framework, and run under a single deterministic entry point that records library versions and digests alongside the results. This is a deliberate property rather than an incidental one: the mathematical claims of Sections 5 and 7 can be checked in full without access to any measured data, and their validity does not rest on either bearing benchmark.

The field experiments necessarily require the CWRU and NASA IMS archives. Both are publicly available, but they are externally maintained and are not redistributed with the package, so reproducing those results involves a separate acquisition step and the training framework used for the shared representation; the corresponding procedure is documented separately. Because the calibration analyses operate on the one-dimensional residual streams alone, they can be repeated from exported residuals without retraining and without redistributing either dataset.

9. Results

Results are reported in the order of Section 8. Throughout, the target coverage is $1 - \alpha = 0.98$, false-alarm rates are out-of-sample, and every quantity produced by the accompanying package is reproducible from the recorded seed.

9.1. Verification of the calibration theory

Exhaustive enumeration. For every combination of $n \in \{4, \dots, 7\}$ and $\alpha \in \{0.2, 0.25, 0.3, 0.4, 0.5\}$ admitted by Corollary 6, enumerating all $(n + 1)!$ orderings reproduces the coverage identity of Proposition 4 exactly, to machine precision, and the rank of the future observation is confirmed equiprobable over $\{1, \dots, n + 1\}$. The feasibility condition and the degenerate-range identity of Corollaries 6 and 7 were checked against their closed forms over a grid of eight values of α spanning 0.005 to 0.25, with agreement at every integer n in a range extending well beyond the boundary in each case. These are checks of the statements themselves rather than of a sampling approximation to them.

The estimator–stopping design. Table 1 reports the four arms across the six residual distributions. Two patterns are visible and neither depends on the distribution.

First, arm A4 attains a pooled coverage of 0.98136 against the value $k_\alpha/(n + 1) = 50/51 = 0.98039$ predicted exactly by Proposition 4, a discrepancy of 1.5 standard errors over 900 trials. The agreement holds for every member of a family chosen to be as dissimilar as the tails, supports and modalities allow, which is the behaviour a distribution-free statement requires and the behaviour whose absence would falsify it.

Second, the two conventions separate cleanly. Decomposing the coverage loss gives -1.87 percentage points attributable to interpolation at fixed n , and -1.84 under adaptive stopping; the effect is stable in sign and magnitude across all six distributions, exactly as Remark 11 predicts. The corresponding entries for the stopping rule are $+0.44$ and $+0.47$ percentage points, that is, in the opposite direction and roughly four times smaller. Interpolation, not adaptive stopping, is the dominant source of the excess false-alarm rate.

Isolating stopping selection from sample size. The comparison just made conflates two effects, because the adaptive rule does not stop at $n = 50$ but at a median of 68 observations, and inside the degenerate range a larger n yields a larger radius. Comparing the adaptive rule against a fixed- n calibration on an independent block of the same realized size isolates the selection effect proper: it is $-0.11, +0.19, +0.01, -0.18, -0.06$ and $+0.00$ percentage

Table 1

Out-of-sample false-alarm rate (%) by arm and residual distribution, nominal level 2.0%. A1 reproduces the published pipeline; A4 is the configuration to which Proposition 4 applies without qualification. The last column is the median commissioning length selected by the adaptive rule in arm A1.

Residual distribution	A1	A2	A3	A4	med. n (A1)
lognormal(0, 0.4)	3.43	1.41	3.74	2.16	96
gamma($k = 3$)	3.09	1.62	3.55	1.81	97
exponential(1)	2.83	1.26	3.72	1.74	119
Pareto($a = 3$)	3.61	1.28	3.87	1.77	118
half-normal	3.14	1.41	3.74	1.89	60
bimodal mixture	3.47	1.54	3.79	1.82	66
pooled coverage (standard error)	0.96737 (0.00119)	0.98580 (0.00058)	0.96264 (0.00043)	0.98136 (0.00063)	
pooled FAR (%)	3.26	1.42	3.74	1.86	

Table 2

Standard deviation of the certified radius against commissioning length, on a lognormal residual stream. Inside the degenerate range of Corollary 7 the estimator is the sample maximum and its dispersion is essentially constant; it contracts only once an interior order statistic becomes available at $n = 99$.

n	k_α	sample max?	sd(τ)	relative to $n = 98$
49	49	yes	0.4999	1.00
60	60	yes	0.5028	1.01
80	80	yes	0.5027	1.01
98	98	yes	0.4987	1.00
99	98	no	0.3061	0.61
120	119	no	0.3081	0.62
200	197	no	0.2012	0.40
400	393	no	0.1340	0.27

points across the six distributions. The magnitude is at most 0.2 points and the sign is not consistent, so the selection mechanism described in Remark 12 is negligible at this configuration and the apparent conservatism of the adaptive arm is a sample-size effect.

Estimator variance and the degenerate range. Table 2 gives the standard deviation of the certified radius as a function of n .

The dispersion is flat to within 1% across [49, 98] and then falls by a factor of 1.6 at the first interior order statistic and by 2.5 by $n = 200$. Clearing the commissioning floor is therefore necessary but not sufficient: a commissioning length chosen just above 49 satisfies Corollary 6 while purchasing the most variable estimator available.

The exchangeability test. Table 3 reports its size and power. The size column lies between 0.025 and 0.083 against a nominal 0.05, within the binomial standard error of 0.0199 at 120 trials, and shows no systematic dependence on the distribution, as a rank-based statistic requires. Power rises monotonically with the injected drift and reaches unity at two standard deviations of accumulated trend in five of six cases. The test is therefore usable as a probe of Assumption 3 rather than as a source of spurious alarms.

9.2. Controlled instantiation on the simulator

End-to-end membership. With the representation operator fitted on 600 nominal windows only and the radius certified from 120 further nominal windows, the held-out false-alarm rate on 1500 nominal windows is 1.13% against a nominal 2%. The median residual rises from 2.32×10^{-3} on nominal windows to 6.81×10^{-3} , 1.03×10^{-2} and 1.65×10^{-2} on ball, outer-race and inner-race faults, and the single membership rule (24) detects them at 79.0%, 99.0% and 100% respectively. No fault observation entered the fitting or the calibration.

Table 3

Rejection rate of the permutation test at level 0.05. The first column is the size, measured on exchangeable streams; the remainder is the power against an injected monotone drift, expressed in standard deviations of accumulated trend.

Residual distribution	size	drift 0.5	drift 1.0	drift 2.0	drift 3.0
lognormal(0, 0.4)	0.042	0.133	0.692	1.000	1.000
gamma($k = 3$)	0.058	0.233	0.692	1.000	1.000
exponential(1)	0.042	0.408	0.825	1.000	1.000
Pareto($a = 3$)	0.025	0.683	0.992	1.000	1.000
half-normal	0.083	0.208	0.592	0.992	1.000
bimodal mixture	0.042	0.525	0.875	1.000	1.000

Table 4

Saturation of the nominal radius. “certified” is the estimator of Corollary 4, “interp” the interpolated empirical quantile; “coverage” and “int. cov” are the corresponding out-of-sample coverages, which the certification requires to be at least $p = 0.98$; “bias” is the relative departure of the mean certified radius from R_p . A dash marks a commissioning length below the floor, at which no radius is certifiable.

n	certified	sd	bias (%)	coverage	interp	int. cov	max?
30	–	–	–	–	21.235	0.9558	–
49	23.176	1.682	+2.13	0.9808	21.496	0.9609	yes
60	23.358	1.685	+2.93	0.9822	21.672	0.9640	yes
80	23.921	1.445	+5.41	0.9878	21.894	0.9690	yes
98	24.152	1.488	+6.43	0.9890	21.998	0.9702	yes
99	22.785	1.290	+0.41	0.9789	21.997	0.9700	no
150	22.677	0.951	–0.07	0.9790	22.115	0.9730	no
300	22.639	0.707	–0.23	0.9793	22.348	0.9762	no
800	22.542	0.398	–0.67	0.9788	22.425	0.9776	no
2000	22.555	0.211	–0.61	0.9789	22.511	0.9783	no

Four simulated assets differing only by an installation signature, sharing one representation operator and one target coverage, identify radii of 7.15×10^{-3} , 7.40×10^{-3} , 1.78×10^{-2} and 1.15×10^{-2} , a spread of 2.49 between the largest and the smallest. This is the behaviour Remark 9 predicts: the representation is shared and its residual scale is not.

Saturation of the nominal radius. Table 4 measures the radius of Definition 3 against a reference $R_p = 22.693$ estimated from 4000 held-out nominal windows.

Three things follow, and the first two are the substance of Remark 5.

Inside the degenerate range the certified radius does not settle: its mean drifts from 2.13% above R_p at $n = 49$ to 6.43% at $n = 98$, and then falls discontinuously to 0.41% at $n = 99$, where an interior order statistic first becomes available. This is not a property of the regime but of the estimator, which is the sample maximum throughout that band by Corollary 7. A saturation curve computed this way cannot flatten there, and apparent settling is the maximum of a growing sample.

The interpolated estimator behaves in the complementary way. It approaches R_p smoothly from below, producing the familiar saturating curve, and its coverage is below the 0.98 target at every commissioning length examined, from 0.956 at $n = 30$ to 0.978 at $n = 2000$. The visual impression of convergence and the coverage guarantee are supplied by different estimators, and not by the same one.

Third, replacing the split-sample reference by a plug-in reference fitted on the same observations shrinks the radius by 2.2 to 3.0% at $n \in \{49, 98, 300, 800\}$, confirming the sign predicted in Remark 3. Coverage, however, is not degraded, because the evaluation distances are recomputed in the same plug-in geometry and the two shift together. The practical cost of a plug-in reference is therefore the comparability of radii across assets and regimes, not the validity of the membership test, and the remark is refined accordingly.

Table 5

Asset-specific reference on CWRU. Each detector is trained and calibrated independently on the complete healthy record of its operating condition. This reference is deliberately favourable and operationally unavailable.

RPM	threshold	healthy FAR (%)	ball DR (%)	inner DR (%)	outer DR (%)
1730	0.6485	2.23	100.00	100.00	100.00
1750	0.7590	2.23	100.00	100.00	100.00
1772	0.7248	2.23	100.00	100.00	100.00
1797	0.4591	2.46	100.00	100.00	100.00
mean	–	2.29 ± 0.12	100.00	100.00	100.00

Table 6

Simulation-to-field transfer on CWRU under one shared representation. Only the information available for estimating the radius differs between strategies. Healthy FAR covers the complete healthy sequence including commissioning observations; post-convergence values appear in Table 7.

RPM	calibration	τ	FAR (%)	ball (%)	inner (%)	outer (%)	windows
1730	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1730	field oracle	0.3578	2.23	99.75	100.00	99.86	–
1730	online	0.3520	3.71	99.75	100.00	100.00	49
1750	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1750	field oracle	0.3434	2.23	100.00	100.00	100.00	–
1750	online	0.3440	1.98	100.00	100.00	100.00	62
1772	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1772	field oracle	0.3512	2.23	100.00	100.00	100.00	–
1772	online	0.3550	1.24	100.00	100.00	100.00	58
1797	direct transfer	0.1027	100.00	100.00	100.00	100.00	–
1797	field oracle	0.3886	2.46	98.03	100.00	100.00	–
1797	online	0.3869	3.45	98.52	100.00	100.00	49

A negative result: the radius is not a detector. Measured under the shared nominal projection, the certified radius of each regime is 23.13 for nominal, 22.63 for outer-race, 22.78 for inner-race and 23.95 for ball faults, that is, ratios to nominal between 0.98 and 1.04. There is no separation. The explanation is mechanical: a two-dimensional principal projection of a raw vibration window is dominated by the shaft harmonics, while the fault signature is a low-amplitude impulse train contributing almost nothing to the leading variance directions. The same simulator, the same nominal-only fitting and the same membership rule detect those faults at 79–100% when the residual is used instead. Saturation of a region and separability of that region from its neighbours are therefore independent properties, as Remark 6 states, and a saturation plot is evidence about the sampling of the nominal geometry rather than about regime discrimination.

9.3. Field transfer: CWRU

The results in this subsection and the next are carried over from the existing pipeline and are not regenerated by the accompanying package, for the reasons given in Section 8.7. By Section 8.6 they were obtained with the interpolated estimator and $N_{\min} = 40$, that is, in the A1 configuration, and therefore do not carry the guarantee of Proposition 4.

Table 5 establishes the reference. Independently trained asset-specific detectors, one representation and one preprocessing transformation per operating condition fitted on that condition's complete healthy record, achieve 100% detection for all three fault families at a mean healthy false-alarm rate of $2.29 \pm 0.12\%$. Their absolute decision thresholds range from 0.459 to 0.759, a spread of 1.65, although their diagnostic operating points are identical. Numerical residual scales are not interchangeable even between models of the same architecture trained on the same data, which is Remark 9 in its sharpest form.

Table 6 reports the three calibration strategies under one shared simulator-trained representation.

Direct transfer of the absolute simulator threshold fails under every operating condition, flagging every healthy window as anomalous while producing an apparent 100% detection rate. This is the predicted outcome of Remark 9

Table 7

Agreement between the converged online radius and the complete-data oracle. Post-calibration FAR is evaluated only on healthy observations occurring after the declared convergence point.

RPM	τ_{oracle}	τ_{online}	rel. diff. (%)	windows	post-cal. FAR (%)
1730	0.3578	0.3520	1.64	49	3.94
1750	0.3434	0.3440	0.19	62	1.75
1772	0.3512	0.3550	1.06	58	0.87
1797	0.3886	0.3869	0.44	49	3.90
mean	—	—	0.83	54.5 ± 6.6	2.62

and is the empirical content of the claim that the representation transfers and its scale does not: the same representation, given a locally estimated radius, recovers an operationally meaningful boundary in the same table.

Local calibration does exactly that. The complete-data field oracle reproduces the expected 2% operating point, and the online procedure reaches a comparable one after 49, 62, 58 and 49 observations, a mean of 54.5 ± 6.6 . Table 7 compares the converged radii with the oracle values.

The converged radii differ from their oracle references by 0.19–1.64%, a mean of 0.83%. The procedure therefore recovers nearly the same acceptance boundary that retrospective access to the complete healthy record would have produced, from at most 62 observations.

The commissioning lengths, and what they do not show. The four commissioning lengths are 49, 62, 58 and 49, and the coverage floor of Corollary 6 at $\alpha = 0.02$ is also 49. By Remark 16 these coincide for unrelated reasons: the stopping rule cannot emit a length below 49 at the configuration used, and that bound does not involve α . The clustering is therefore uninformative about the floor, and is reported here only to forestall the inference.

More consequentially, all four lengths lie in [49, 98], which by Corollary 7 is the range in which a certified radius would be the sample maximum. Combined with the use of an interpolated quantile, this places the field calibration in precisely the regime that Table 4 characterizes as simultaneously the most variable and the least conservative.

Robustness, read as a probe of exchangeability. Permuting the commissioning observations and their arrival order over five realizations per condition, with the representation and all parameters fixed, gives 20 convergent runs out of 20. The estimated radius stays within $2.08 \pm 0.71\%$ of the oracle and the fleet-averaged post-calibration false-alarm rate is $3.59 \pm 1.59\%$. Following Remark 13, the absence of systematic dependence on arrival order is the evidence bearing on Assumption 3, and it is consistent with the assumption holding within the commissioning window.

The dispersion is nonetheless substantial: one 1750 RPM realization produced a post-calibration false-alarm rate of 13.8% while remaining within 4% of the oracle radius. This is finite-sample uncertainty in an extreme quantile estimated inside the degenerate range, and Table 2 gives its scale directly: at $n \approx 55$ the estimator has the same dispersion as at $n = 49$, and 2.5 times that available at $n = 200$.

Cost of sharing one representation. Relative to the asset-specific reference of Table 5, replacing four independently trained detectors by one simulator-trained representation with local calibration changes mean ball-fault detection from $100.00 \pm 0.00\%$ to $99.57 \pm 0.71\%$, leaves inner- and outer-race detection at 100%, and raises the mean healthy false-alarm rate from $2.29 \pm 0.12\%$ to $2.60 \pm 1.18\%$. The reference requires a fitted preprocessing transformation, a trained representation and the complete healthy record of every condition; the shared alternative requires 49–62 nominal observations and no field training.

9.4. Field transfer: NASA IMS

Table 8 reports the four run-to-failure bearings.

Calibration converged for all four bearings after 49–57 recordings. The converged radii span 6.67×10^{-3} to 1.89×10^{-2} , a factor of 2.8, under an identical representation, an identical target coverage and an identical stopping rule. This is the same phenomenon as the simulated assets of Section 9.2 and as the CWRU threshold spread of Table 5, now on physically distinct units: the transferable object is the representation, and the radius is individual. Corollary 2 is the structural statement behind it, applied per physical unit rather than per operating condition.

Table 8

NASA IMS deployment under one shared simulator-trained representation with local radius estimation. Relative error is against the complete-data reference for that bearing; lead time is measured to the documented end of the experiment.

Bearing	windows	τ_k	rel. error (%)	post-dep. exceedance (%)	lead time (h)
1	49	0.010680	0.10	99.12	74.00
2	57	0.014950	0.35	86.42	38.83
3	49	0.018930	2.87	95.88	14.50
4	49	0.006672	0.18	97.02	54.33
mean	51	–	0.87	94.61	45.42

All four bearings subsequently departed persistently from their frozen neighbourhood, with a mean post-departure exceedance of 94.61% and lead times of 14.5 to 74.0 hours, a mean of 45.4. Repeating the commissioning with five permuted early-life samples per bearing gave 20 convergent runs out of 20, radii within 0.83% of their references on average, and a healthy false-alarm rate of 4.98%; bearing 3, which shows the largest deviation in Table 8, is also the most sensitive to commissioning-sample variability, at $1.58 \pm 1.88\%$.

Two caveats attach to the lead times specifically. They depend on the persistence rule and on the recording-level aggregation, and Section 8.7 records that the source code and the accompanying text disagree on both: three exceedances in five recordings against eight in ten, and a mean against a 95th-percentile aggregation. The values above are those produced by the code as executed. Resolving the discrepancy is a prerequisite for quoting lead times as a headline result, and is listed in Section 12.

9.5. Summary

The verification experiments confirm Proposition 4 exactly by enumeration and to within 1.5 standard errors by simulation across six dissimilar residual distributions, and they identify interpolation rather than adaptive stopping as the dominant source of the excess false-alarm rate observed in practice. The simulator experiments exercise the full chain and supply two results that qualify how saturation evidence should be read: the certified radius cannot flatten inside the degenerate range, and the radius does not discriminate between regimes that the residual separates almost perfectly. The field experiments show that the representation transfers while its numerical scale does not, that local calibration recovers the complete-data operating point from at most 62 observations, and that independently commissioned physical units identify radii differing by a factor of 2.8 under one shared representation. The field calibration, however, was performed with the estimator and at the commissioning lengths that Sections 7.4 and 8.6 identify as uncertified and maximally variable; re-running it is the single most valuable extension of this work.

10. Discussion

10.1. What the quotient construction delivers, and what it does not

The results are best read as an asymmetry. Four things follow from the geometry, and one conspicuously does not.

Diagnosis factors through the quotient (Proposition 2), so a correct diagnostic decision exists as a function of the equivalence class alone; this is what licenses seeking a representation at all rather than a per-asset classifier. The nominal region is compact and hence has a finite radius (Proposition 1, Corollary 3), so there is a well-defined finite quantity to estimate, which is not automatic and does not follow from the informal saturation criterion on its own. That radius admits a distribution-free estimate from a computable number of observations (Proposition 4, Corollary 6), which converts an engineering threshold into a certified quantity. And regions once recovered can be labelled and calibrated individually with per-class validity (Corollary 5), so the taxonomy can grow after deployment without recalibrating what already exists.

What does not follow is separation. Nothing in the construction bounds the probability of *admitting* a competing region, and Remark 7 shows the guarantee is fully compatible with a permanently ambiguous diagnosis. The framework is therefore a theory of detection with a placeholder where classification should be, and the placeholder is structural rather than an artefact of the particular operator: a scalar residual is many-to-one and discards the direction of departure, and the empirical result of Section 9.2 shows that the low-dimensional radius which describes the nominal geometry

does not discriminate the regimes it describes. Stating this plainly seems preferable to presenting an arg min over regions as though it inherited the guarantee that the membership test does.

10.2. The transferable object is the representation, not its scale

The claim that a representation transfers while its residual scale does not now rests on three independent observations of increasing force.

Simulated assets differing only by an installation signature identify radii spanning a factor of 2.49 under one operator. Physically distinct NASA IMS bearings, independently commissioned under one representation and one convergence criterion, identify radii spanning a factor of 2.8. The sharpest case, however, is the CWRU reference of Table 5: four detectors of identical architecture, trained on the data of their own operating condition, produce decision thresholds from 0.459 to 0.759 while achieving indistinguishable diagnostic operating points. Here the variation cannot be attributed to the asset, the sensor or the operating point, because it survives when those are held fixed and only the optimization trajectory differs.

The conclusion is stronger than a deployment convenience. Diagnostic performance is a property of the learned geometry; the numerical value of the residual is a property of a particular fit. Transferring the former and re-estimating the latter is therefore not a heuristic for saving effort but the only combination consistent with what the residual actually is, and Remark 9 gives the reason: the membership predicate inherits orbit-invariance from Proposition 2, while the score used to evaluate it does not.

10.3. Saturation, re-read

Three findings revise how the saturation evidence of the companion work [4] should be interpreted, and the revisions follow from proofs rather than from anything discovered in the data.

The first concerns the estimator. Between $1/\alpha - 1$ and $2/\alpha - 1$ the certified radius is the sample maximum (Corollary 7), which increases with n ; the measured drift is from +2.13% to +6.43% of R_p across that band, followed by a discontinuous fall. A curve computed this way cannot flatten inside the band, so flattening observed there is not evidence that the geometry has stabilized.

The second concerns which estimator produces the familiar picture. The interpolated quantile approaches R_p smoothly from below and yields the reassuring saturating curve, and it undercovers at every commissioning length examined. The visual and the guarantee are supplied by different estimators. This is not a reason to abandon saturation plots, but it is a reason to report attained coverage beside them, since the plot alone cannot distinguish a saturated regime from a conveniently biased statistic.

The third is the negative result. Fault regimes measured under the shared nominal projection have radii within 4% of nominal, while the residual detects the same faults at 79–100%. Saturation characterizes how well a region has been sampled; it says nothing about whether that region is distinguishable from its neighbours. Evidence of the first has occasionally been offered in support of the second, including in the framing that motivated this work, and the two should be kept apart.

10.4. Relation to conformal prediction

The calibration machinery is a distribution-free tolerance limit in the sense of Wilks [15], and the multi-region rule (17) is the class-conditional form of a conformal predictor [11, 14]. It is worth being explicit that this was arrived at from the quotient geometry rather than imported: the radius is what the compactness result says exists, and exchangeability is what the nominal regime supplies during commissioning.

The connection is useful in both directions. It supplies the guarantee at essentially no cost, since the estimator is an order statistic of quantities the pipeline already computes. It also identifies precisely where the present framework is weak, because the gap named in Remark 7 is the distinction between validity and efficiency that conformal prediction has studied at length: a predictor can be valid and useless if its prediction sets are large. Read this way, the classification problem left open in Section 6 is not an unexplored question but a known one, and the literature on conditional validity and on efficient non-conformity scores is the natural place to look for a construction that tightens $\Gamma(x)$ without sacrificing Corollary 5.

10.5. Consequences for practice

Five recommendations follow, each from a proved statement rather than from tuning.

1. Calibrate with the order statistic $r_{(k_\alpha)}$ rather than with an interpolated empirical quantile. The interpolated value lies at or below it and does not satisfy the coverage bound; the measured cost is 1.9 percentage points of coverage, consistently in sign across every distribution examined.
2. Require $n \geq 2/\alpha - 1$, not merely $n \geq 1/\alpha - 1$. Clearing the floor makes a radius certifiable; clearing the degenerate range makes it stable, and the difference is a factor of 2.5 in the estimator's dispersion.
3. Report attained coverage alongside any saturation curve, for the reason given in Section 10.3.
4. Treat permutation of commissioning order as a test of Assumption 3 rather than as a robustness check. The test has nominal size and high power against drift, so a systematic dependence on arrival order would be informative rather than merely inconvenient.
5. Estimate the reference point, and any projection, on observations disjoint from those used to compute the radius. The measured plug-in shrinkage is 2–3%; coverage survives, but radii computed in different plug-in geometries are not comparable across assets, which is what a fleet-level view requires.

None of these requires additional data, additional labels, or a different architecture.

10.6. A quantitative reading of the governing thesis

The premise stated in Section 1 is that physically generated signals do not exhibit unbounded or adversarial variation, and that departure from a compact nominal region is what an anomaly is. The results give that premise two quantitative renderings, one structural and one operational, and both are falsifiable.

The structural rendering is Corollary 3: under bounded nuisance the nominal orbit is relatively compact, so the radius is finite and there is something to estimate. Were nominal variation genuinely unbounded, no finite radius would exist and the entire construction would be vacuous rather than merely inaccurate.

The operational rendering is the commissioning floor. At a target coverage of 98%, 49 nominal observations suffice to certify a boundary distribution-free, and 99 suffice to do so with a stable estimator. That a regime can be bounded from tens of observations is the concrete content of the claim that physical variability is compact; a process without that structure would require the radius to keep growing with n , which is exactly what Table 4 shows the estimator doing inside the degenerate range and not doing outside it.

The NASA IMS trajectories close the loop on the diagnostic side. Four independently commissioned bearings, with no fault labels at any stage, departed persistently from their own frozen neighbourhoods with lead times of 14.5 to 74.0 hours. The anomaly signal is not a learned fault signature but the failure of membership in a region whose extent was estimated from roughly fifty nominal observations, which is the inverse-problem claim of this paper reduced to its operational minimum.

11. Limitations

12. Future Work

13. Conclusion

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